

---

# **Software Requirements Specification**

**for**

## **Roadmap planner**

**Version 1.0 approved**

**Prepared by Team 14(Younhong Jeon, Raehyun Kim, Jimin Seo,  
Hyunho Na)**

**KAIST**

**April 1, 2026**

# Table of Contents

|                                               |           |
|-----------------------------------------------|-----------|
| <b>1. Introduction</b>                        | <b>1</b>  |
| 1.1 Purpose                                   | 1         |
| 1.2 Document Conventions                      | 1         |
| 1.3 Intended Audience and Reading Suggestions | 1         |
| 1.4 Product Scope                             | 2         |
| 1.5 References                                | 2         |
| <b>2. Overall Description</b>                 | <b>3</b>  |
| 2.1 Product Perspective                       | 3         |
| 2.2 Product Functions                         | 3         |
| 2.3 User Classes and Characteristics          | 4         |
| 2.4 Operating Environment                     | 4         |
| 2.5 Design and Implementation Constraints     | 4         |
| 2.6 User Documentation                        | 5         |
| 2.7 Assumptions and Dependencies              | 5         |
| <b>3. External Interface Requirements</b>     | <b>7</b>  |
| 3.1 User Interfaces                           | 7         |
| 3.2 Hardware Interfaces                       | 9         |
| 3.3 Software Interfaces                       | 9         |
| 3.4 Communications Interfaces                 | 10        |
| <b>4. System Features</b>                     | <b>11</b> |
| 4.1 Credit and GPA Calculator                 | 11        |
| 4.2 Course Prerequisite Tree                  | 15        |
| 4.3 Semester Roadmap                          | 19        |
| 4.4 Sidebar                                   | 24        |
| 4.5 Settings                                  | 28        |
| 4.6 Authentication                            | 30        |
| <b>5. Other Nonfunctional Requirements</b>    | <b>31</b> |
| 5.1 Performance Requirements                  | 31        |
| 5.2 Safety Requirements                       | 31        |
| 5.3 Security Requirements                     | 31        |
| 5.4 Software Quality Attributes               | 32        |
| 5.5 Business Rules                            | 32        |
| <b>6. Other Requirements</b>                  | <b>33</b> |
| Appendix A: Glossary                          | 33        |
| Appendix B: Analysis Models                   | 34        |
| Appendix C: To Be Determined List             | 41        |

## Revision History

| Name          | Date       | Reason For Changes | Version |
|---------------|------------|--------------------|---------|
| Initial Draft | 2026-04-01 | Initial Draft      | 1.0     |
|               |            |                    |         |

# **1. Introduction**

## **1.1 Purpose**

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the KAIST CS Roadmap Planner — an app-based academic planning application for students enrolled in the School of Computing at KAIST. The document specifies requirements for version 1.0 of the product.

The planner assists students in tracking credit completion, visualizing course prerequisites as an interactive tree, building a semester-by-semester roadmap, and monitoring graduation requirements specific to the KAIST CS undergraduate curriculum.

## **1.2 Document Conventions**

This document is written using the following term.

- KAIST CS Roadmap Planner - Full service name
- Course Prerequisite Tree - Visualization and course selection tool for prerequisites
- Semester Roadmap - Grid-based academic planning tool
- Credit and GPA Calculator - Module for graduation requirements and GPA calculation
- Category - Categories of courses, such as General Required or Data Science
- General Required - General required courses (기초필수)
- General Elective - General elective courses (기초선택)
- Major Required - Major required courses (전공필수)
- Major Elective - Major elective courses (전공선택)
- Completed - Course status for earned credits with grades
- Not Taken - Course status for subjects not yet enrolled
- Planned - Course status for selected subjects
- Custom - User-defined course entries for credits not covered by standard categories.

## **1.3 Intended Audience and Reading Suggestions**

This document is intended for the development team that has adopted this Software Requirements Specification (SRS) for the development of the KAIST CS Roadmap Planner. The team includes project managers, developers, stakeholders, and testers.

The suggested review points for each role are as follows:

- Developers: Read all sections, with emphasis on Sections 3, 4, and 5.
- Project Managers: Sections 1, 2, and 5 provide scope and constraints.

- KAIST Academic Affairs / Stakeholders: Sections 2 and 4 describe product functionality.
- QA / Testers: Section 4 system features with all REQ entries form the test basis.

## **1.4 Product Scope**

The KAIST CS Roadmap Planner is a responsive application that provides KAIST School of Computing students with:

- Credit and GPA Calculator: Features a tracking system for Major Required, Major Elective, and graduation research, allowing students to monitor their progress through category-specific filtering.
- Interactive Course Prerequisite Tree: Offers a visual, hierarchical representation of course dependencies. Users can filter the tree by sub-disciplines to identify optimal learning paths.
- Semester Roadmap: A long-term planning tool that enables users to visually organize their academic schedule across multiple semesters using an intuitive drag-and-drop interface.
- KAIST SSO Integration: Provides secure and seamless access by integrating with the university's Single Sign-On (SSO) authentication system.

The product aims to reduce student confusion around graduation requirements, course sequencing, and semester workload planning.

## **1.5 References**

KAIST School of Computing Undergraduate Curriculum Guide (2024)

KAIST OTL (Online Timetable) — <https://otl.kaist.ac.kr>

IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications

KAIST Graduation Requirements for Computer Science B.S. (2024)

KAIST Academic Policies: Course Registration — official KAIST academic registration policy and schedule.

## 2. Overall Description

### 2.1 Product Perspective

The KAIST CS Roadmap Planner is a new standalone product designed to support academic planning for students in the KAIST School of Computing. The product does not replace existing KAIST systems (e.g., ZEUS, OTL), but rather complements them by providing a student-centered planning layer.

- At launch, course data and graduation requirements will be maintained as a curated internal dataset. Future versions may integrate with KAIST academic APIs for automated data synchronization. The system consists of six major modules that interact closely:
- Login — Handles KAIST SSO authentication and session management
- Credit and GPA Calculator — Provides credit tracking and GPA statistics, integrated with the Semester Roadmap
- Course Prerequisite Tree — Represents the hierarchical prerequisite structure of courses, linked to user progress
- Semester Roadmap — Enables drag-and-drop semester planning across academic years
- Sidebar — Allows users to search and find detailed course data.
- Settings — Manages user preferences such as theme and major/minor selection

### 2.2 Product Functions

At a high level, the product provides the following major functions:

- Display earned and required credits per category (e.g., General Required, General Elective, Major Required, Major Elective, graduation research, custom categories)
- Calculate and present cumulative GPA as well as category-specific GPA
- Provide current semester statistics and projected credits based on Planned courses
- Visualize the course structure as a collapsible Course Prerequisite Tree grouped by academic domains
- Allow users to mark courses as Completed, in-progress, or Planned
- Generate and manage a drag-and-drop Semester Roadmap spanning all academic years
- Support switching between major and minor (or double major) views via Settings

- Offer course search with categories
- Provide user interface customization (e.g., dark/light mode, Settings) and KAIST SSO-based login

## 2.3 User Classes and Characteristics

The primary user classes of the system are defined based on academic progression within the School of Computing:

| User Class                   | Characteristics                                                                                  | Primary Modules Used                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Freshman / Sophomore         | New to CS curriculum; exploring Course Prerequisite Tree and planning early course selections    | Course Prerequisite Tree, Semester Roadmap                            |
| Junior / Senior              | Tracking graduation requirements; optimizing Major Elective / General Elective selection and GPA | Credit and GPA Calculator, Course Prerequisite Tree, Semester Roadmap |
| Double Major / Minor Student | Needs to track requirements across two programs simultaneously                                   | Credit and GPA Calculator, Semester Roadmap                           |

## 2.4 Operating Environment

The system operates in the following environment:

- **Client:** Mobile application
- **Target Devices:** Smartphones (iOS and Android), with support for various screen sizes
- **Framework:** Cross-platform development (for example, using React Native)
- **Authentication:** KAIST SSO (SAML or OAuth2 via KAIST Identity Provider)
- **Backend:** Linux-based server environment providing RESTful APIs
- **Network:** Internet connection required for authentication and data synchronization
- **Offline Support:** Limited read-only access to cached data is available; data modification requires an active network connection

## 2.5 Design and Implementation Constraints

The Product must satisfy the following constraints:

- **Authentication Constraint:** The system shall use KAIST SSO for authentication. If the SSO service is unavailable, authentication via official KAIST email shall be permitted. No other authentication methods or separate credential storage shall be allowed.
- **Data Accuracy Constraint:** All course information and graduation requirements must accurately reflect the official KAIST School of Computing undergraduate curriculum.
- **Localization Constraint:** The product must support both Korean and English user interfaces.
- **Data Update Constraint:** Course data and graduation requirements must be updatable without requiring full product redeployment (e.g., via backend configuration).
- **UI/UX Constraint:** The product must handle complex hierarchical structures (Course Prerequisite Tree) within the limitations of mobile interfaces, ensuring usability and readability.
- **Interaction Constraint:** Drag-and-drop interactions must be supported on touch-based devices with smooth and responsive performance.
- **Performance Constraint:** Real-time calculations (e.g., GPA, credit updates) must be performed with minimal latency to maintain a seamless user experience.
- **Security Constraint:** All user data must be securely transmitted and stored, following standard security practices (e.g., HTTPS, secure token handling).

## 2.6 User Documentation

The Product will provide the following user documentation:

- In-product onboarding walkthrough for first-time users
- Contextual tooltips for major interface elements
- Help page explaining course category abbreviations (e.g., General Required, General Elective, Major Required, Major Elective, Graduation Research)
- FAQ section covering topics such as graduation research and alternative credit recognition

## 2.7 Assumptions and Dependencies

The Product is developed under the following assumptions:

- KAIST SSO service is available and accessible to all enrolled students.



- Official graduation requirements for the School of Computing remain stable within an academic year.

*The Product depends on the following:*

- Manually curated Course Prerequisite Tree and curriculum data at initial release.
- Potential future integration with KAIST academic systems or external APIs.

## **3. External Interface Requirements**

### **3.1 User Interfaces**

#### **1) Login Page**

The Login Page will allow users to access the product through KAIST SSO. Since the product is intended for KAIST students who need to manage their Semester Roadmap based on official academic requirements, the Product should provide a login interface connected to KAIST SSO or an equivalent institutional authentication process. After successful authentication, the user should be redirected to the main page of the product. If authentication fails, the Product should display an appropriate error message and remain on the login page. A loading page may be shown while authentication or initial data loading is in progress.

#### **2) Curriculum Tree Page**

The Course Prerequisite Tree will serve as the main page of the product. It will provide a visual roadmap of the School of Computing curriculum so that users can understand prerequisite relations, recommended order of courses, and the overall structure of the degree. Since the full curriculum may become difficult to view at once, the page should support categorized views so that users can focus on a specific field such as AI, Systems and Networks, Theory, Software Design, or other tracks.

The Product should allow users to reflect Completed courses so that the tree can distinguish Completed courses from remaining courses. When a user selects a course, the Product should show summary information such as course code, title, credit value, and rough course level.

The Course Prerequisite Tree should support scrolling, zooming, and expandable category sections because the curriculum tree may be too large for a single mobile screen. When users select or check courses on this page, the product should update the counts related to Credit and GPA Calculator in real time.

#### **3) Semester Roadmap Page**

The Semester Roadmap Page will allow users to distribute courses across semesters and build a long-term graduation plan. The page should initially support an eight-semester roadmap, while still allowing additional semesters if needed. Since required courses may not always be taken in the originally recommended semester, the system should allow flexible rearrangement of courses across future semesters.

The main interaction on this page should be drag-and-drop based. Users should be able to move courses from the curriculum tree or recommendation area into semester boxes and rearrange them later. Each semester section should display the list of selected courses and the total number of planned credits. The system should count credits in real time. If a course is placed before its prerequisite or in an inappropriate semester, the system should display a warning message or recommend moving it to a later semester.

The page should display roadmap data according to the academic option selected in the Settings Page. In the current version, this page is intended to support semester-level planning rather than a detailed weekly timetable.

#### **4) Status and GPA Calculator Page**

The Credit and GPA Calculator Page will summarize the user's graduation progress and provide credit and GPA calculations. The upper part of the page should present key statistics such as Completed credits, remaining credits, requirement progress, and average GPA. Graduation requirements should be shown by category, such as Major Required, Major Elective, and other requirement groups. The page should support filtered views so that users can examine only a selected category.

The page should also support selection of the current semester for statistics. In addition, the interface may indicate the amount of credits Planned for the next semester in a concise format such as "(+number)." Graduation research progress should be displayed as a separate item when applicable. Courses that are Not Taken may be treated with a default status such as "not counted" or "Not Taken" until the user specifies further information.

The lower part of the page may separately show detailed course lists, requirement breakdowns, and GPA-related information. The interface should clearly distinguish Completed, Not Taken, and Planned items.

#### **5) Settings Page**

The Settings Page will allow users to configure general application preferences and basic academic options. The page should provide toggle controls for sound and vibration settings, allowing users to adjust simple feedback-related preferences within the application. In addition, the page should include an Academic Settings section where users can select their academic option, such as a minor, so that the system can reflect the user's curriculum structure more appropriately. Although this page is not a core academic feature, it supports personalization and improves the overall usability and consistency of the application.

#### **6) Common Navigation**

Since the application contains several major pages, users should be able to move among the Curriculum Tree Page, Semester Roadmap Page, Status and GPA Calculator Page, Settings Page, and Login Page through a simple and consistent navigation structure. A bottom navigation bar or tab-based navigation is recommended for the major pages of the service. Error messages, warnings, and confirmation messages should be shown in a clear and understandable manner.

## 7) Sidebar

The Sidebar is a global UI component that appears as a small icon in the top right corner of any tab. Upon clicking the icon, the Sidebar expands to reveal various categories and a search bar at the top. Users can search for courses either by selecting a category or by entering text into the search bar within the expanded interface.

When a specific course is selected from the search results, the system displays detailed course information along with two functional buttons. These include the "Find in Tree" feature, which locates the course within the Course Prerequisite Tree, and the "Add to Selected Courses" feature, which allows users to add the course to their planned Courses.

## 3.2 Hardware Interfaces

The KAIST CS Roadmap Planner is primarily designed for smartphones and tablets. The main hardware interface of the system is the touch screen, through which users will tap, scroll, zoom, search, and drag courses across the Semester Roadmap. Because the Course Prerequisite Tree may contain a large amount of visual information, the application should also remain usable on larger tablet screens.

The system is intended to support modern Android and iOS mobile devices. A recommended baseline is Android 9 or higher with SDK version 28 or higher, and iOS 15 or higher. No specialized external hardware is required beyond a device capable of running the application and connecting to the Internet.

## 3.3 Software Interfaces

### 1) Backend Server

The backend server will provide the core logic and data required by the application. It should provide APIs for retrieving Course Prerequisite Tree data, course information, graduation requirement data, Semester Roadmap records, and Credit and GPA Calculator results. The backend should also validate user input and synchronize changes across multiple pages.

For Login, the backend should receive the user's KAIST ID and password from the frontend, call the KAIST Login API or an equivalent institutional Login interface, obtain a Login token, and store the token securely for Login session and expiration management. The backend should manage Login validity and re-Login when necessary.

The backend should also store and manage user-specific academic planning data, including Completed courses, checked or Planned courses, and roadmap or timetable-related information. The server may be deployed in a cloud environment such as AWS or an equivalent platform.

## **2) Database**

The database will store course metadata, curriculum structures, user-selected academic options, Completed courses, Planned courses, Semester Roadmap records, timetable-related information, and Credit and GPA Calculator data. Whenever a user adds, removes, or rearranges a course, the system should store the updated plan and reflect the change in the Credit and GPA Calculator Page.

A MongoDB-based NoSQL database is recommended for flexible management of curriculum structures and user planning data, although other database solutions may also be used if they satisfy the system requirements.

## **3) Academic Data Interface**

Because graduation requirements and course classifications may change over time, the system should support an interface through which official academic data can be updated. This may be implemented through an administrator-managed update system or another trusted academic data source. Updated information should be reflected consistently across the Course Prerequisite Tree, Semester Roadmap, and Credit and GPA Calculator.

## **4) Authentication Interface**

Since the service is intended for KAIST students, the system should interface with Login or an equivalent institutional Login service. This interface should verify the user before granting access to saved academic plans and personal progress information.

## **3.4 Communications Interfaces**

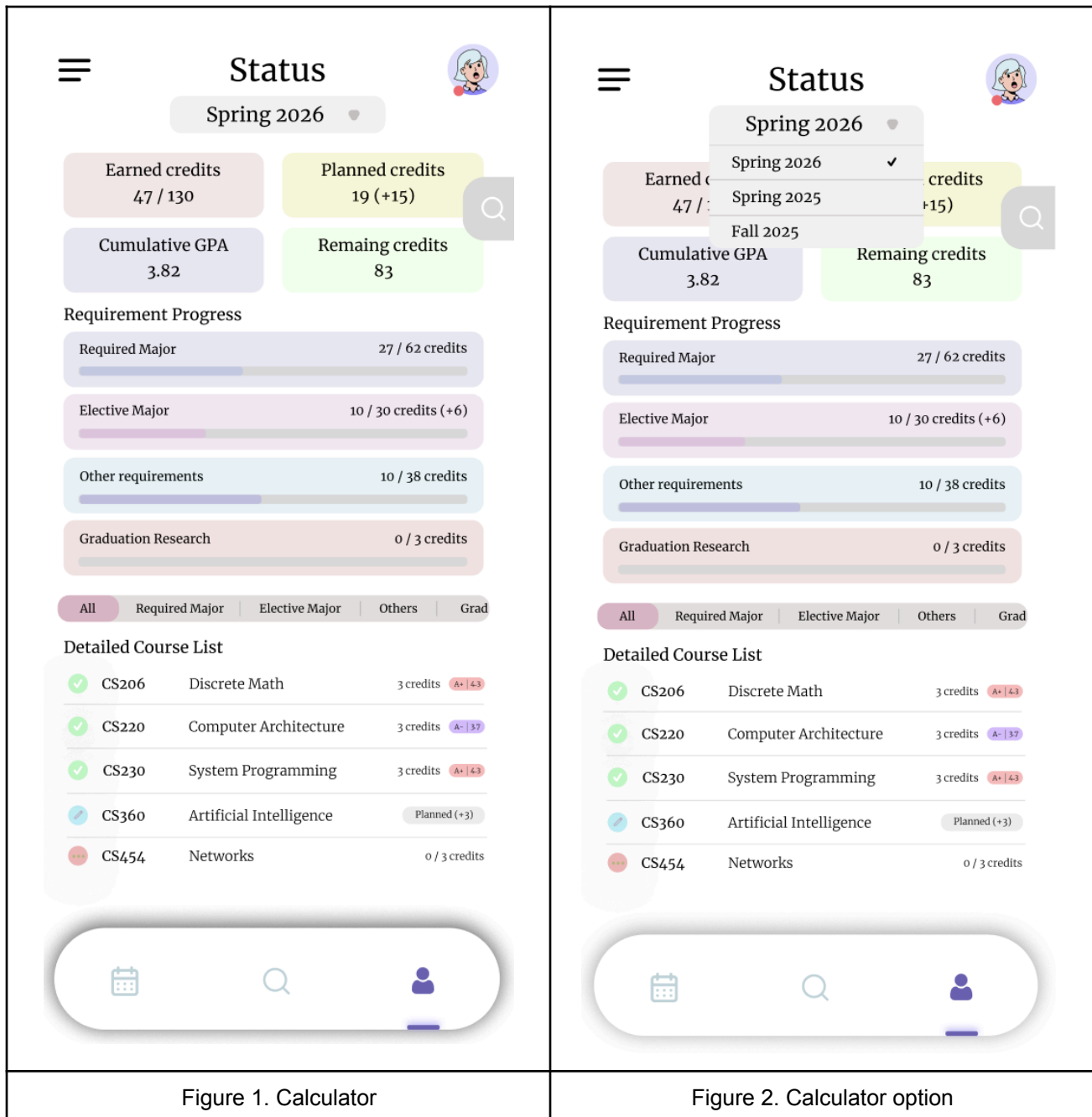
The application should communicate with the backend server through the Internet. The recommended protocol is HTTPS in order to protect personal information and academic planning data. Data exchanged between the client and the server should be formatted in JSON for compatibility and simplicity. Typical communication includes Login requests, retrieval of Course Prerequisite Tree data, course-detail requests, Semester Roadmap updates, Completed status updates, and Credit and GPA Calculator results. Since the current version of the service does not require real-time chat or collaboration, a standard request-response communication model is sufficient. REST-style API communication is recommended for the current version of the application.

## 4. System Features

The system is organized around four primary feature modules and two more modules. Each module's requirements are listed below in REQ tables.

### 4.1 Credit and GPA Calculator

#### 4.1.1 Description and Priority



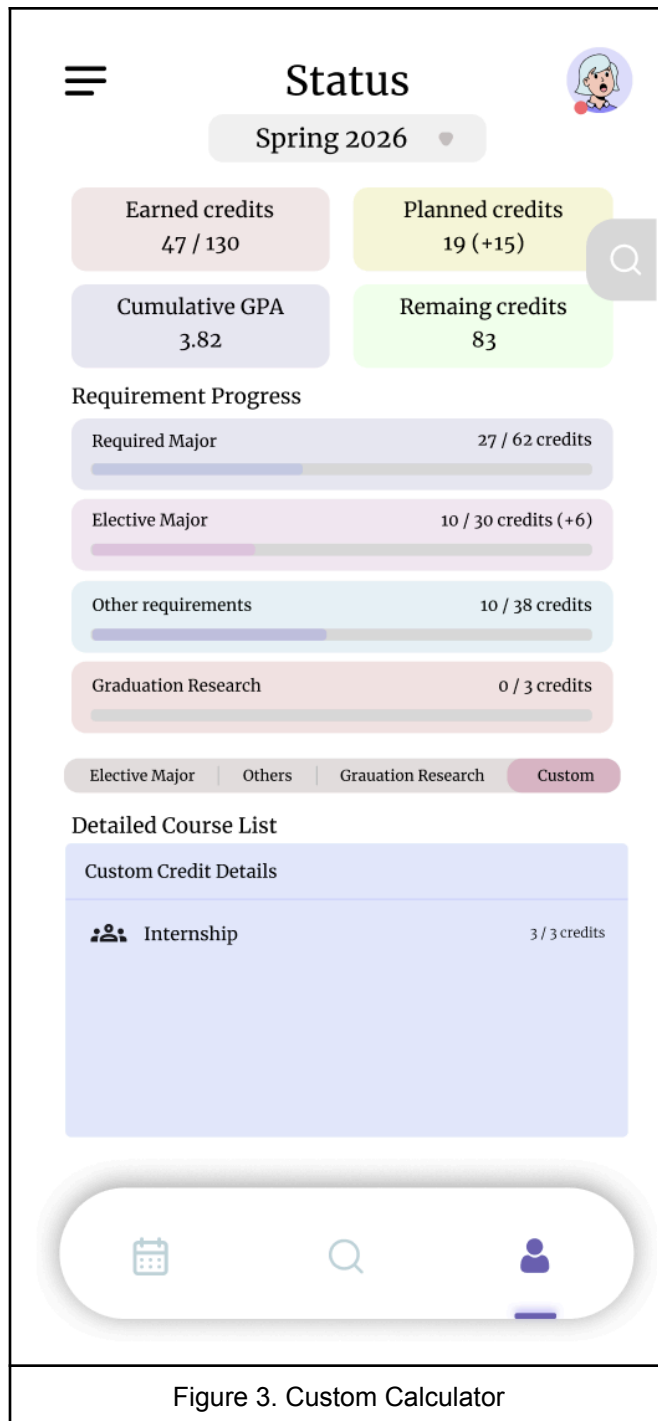


Figure 3. Custom Calculator

The Credit and GPA Calculator is a core module of the KAIST CS Roadmap Planner designed to provide an integrated overview of a user's graduation requirement status and academic performance. This module calculates the Cumulative GPA and major-specific GPA based on the grades of Completed courses. It maintains real-time synchronization with the semester roadmap to visualize how Planned credits impact the overall academic trajectory. Users can intuitively identify missing requirements through categorized progress bars for General Required, General Elective, Major Required, and Major Elective subjects. Furthermore, it interfaces with the Course Prerequisite Tree to ensure that credit calculations are based on a valid sequence of subjects. This is a high-priority core feature.

#### 4.1.2 Stimulus/Response Sequences

- User opens Credit and GPA Calculator tab → System displays credit summary panel with category breakdown and filtered course list
- User selects a category filter (e.g., Major Required) → System updates the course list and recalculates stats for that category only
- User inputs or changes GPA for a course → System immediately recalculates cumulative and category GPA
- User marks a course as Not Taken → System excludes it from GPA calculations
- System displays '(+N credits)' projection for Planned courses

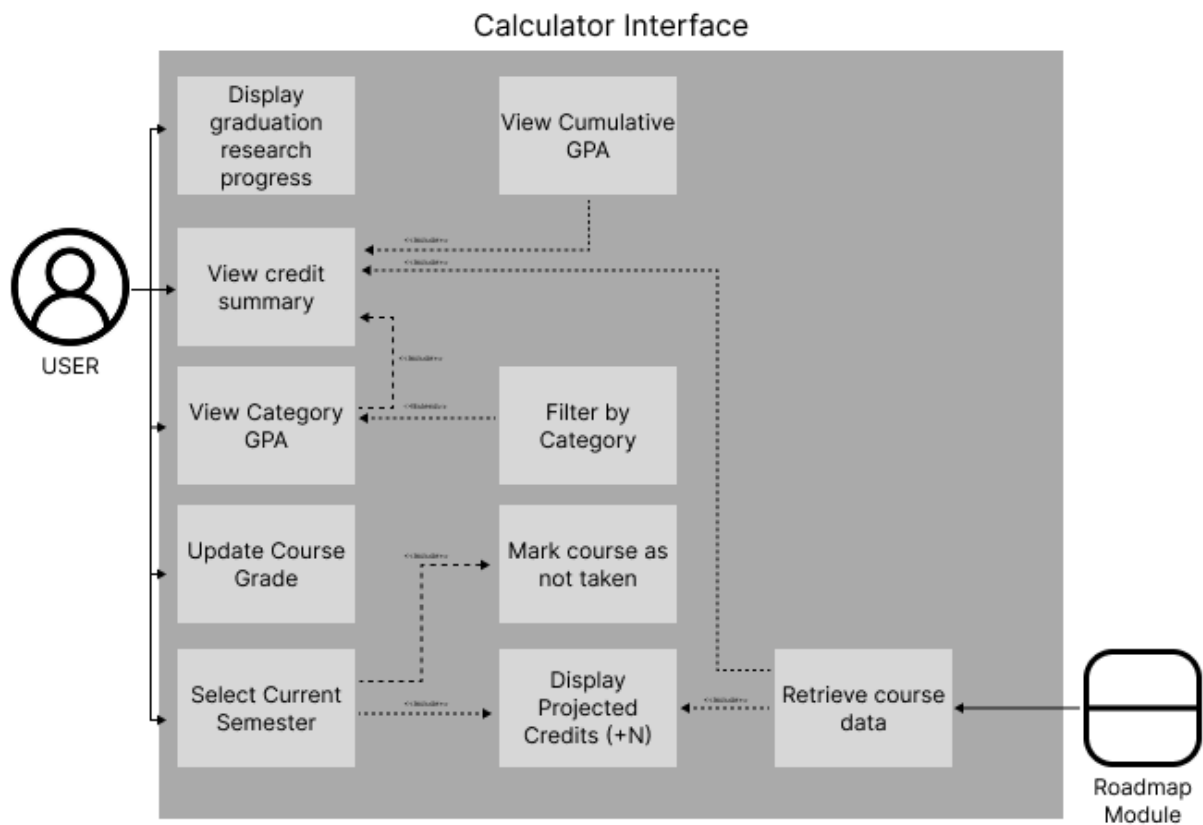
#### 4.1.3 Functional Requirements

| ID                  | Requirement Description                                                                                                                                                                                                                      | Priority |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>REQ-CALC-001</b> | The system shall display Completed credits vs. required credits for each course category: General Required, General Elective, Major Required, Major Elective, Graduation Research, and Custom.                                               | High     |
| <b>REQ-CALC-002</b> | The system shall display a cumulative GPA and a per-category GPA in the summary stats panel.                                                                                                                                                 | High     |
| <b>REQ-CALC-003</b> | The system shall provide a category filter control allowing the user to view course lists and stats for a single selected category.                                                                                                          | High     |
| <b>REQ-CALC-004</b> | Each course entry in the list shall have a GPA input field. The default state for each course shall be "Not Taken", excluding it from GPA calculations.                                                                                      | Medium   |
| <b>REQ-CALC-005</b> | The system shall display a current-semester selector in the stats panel. When a semester is selected, Planned courses for that semester (from the Semester Roadmap) shall be reflected as a '(+N credits)' addition to the displayed totals. | Medium   |
| <b>REQ-CALC-006</b> | Graduation Research progress shall be displayed as a fraction, e.g., '0/3', in the stats panel.                                                                                                                                              | High     |
| <b>REQ-CALC-007</b> | Courses marked as Custom (user-added) shall appear in the course list without any special badge or annotation.                                                                                                                               | Medium   |
| <b>REQ-CALC-008</b> | The Credit and GPA Calculator course list shall be synchronized with the Semester Roadmap — courses added to the Semester Roadmap shall automatically appear in the Credit and GPA Calculator list.                                          | High     |
| <b>REQ-CALC-009</b> | GPA calculations shall use the KAIST 4.3 grading scale (A+: 4.3, A0: 4.0, B+: 3.3, ..., F: 0.0).                                                                                                                                             | High     |



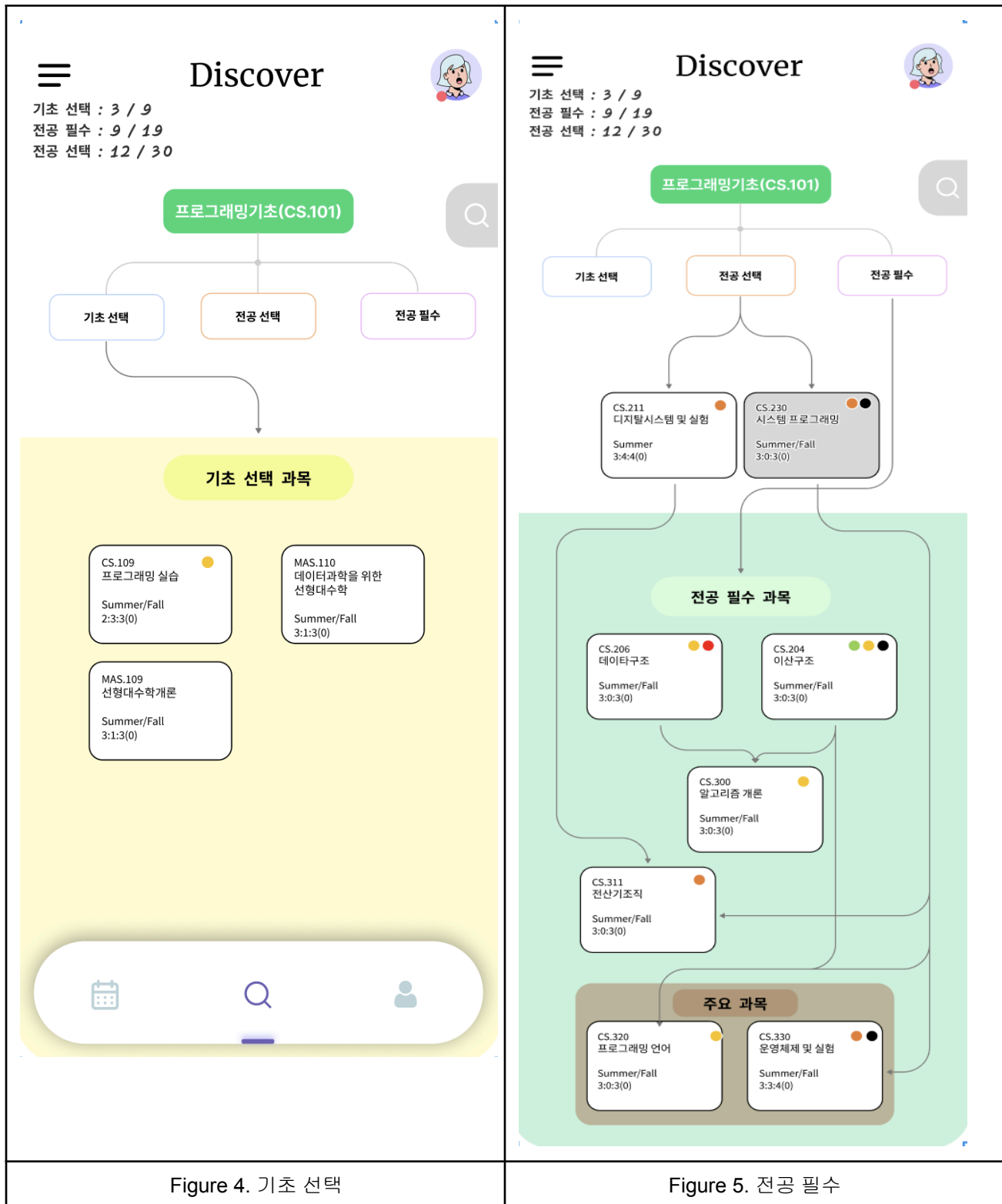
| ID                  | Requirement Description                                                                        | Priority |
|---------------------|------------------------------------------------------------------------------------------------|----------|
| <b>REQ-CALC-010</b> | The system shall display a loading state indicator while recalculating stats after user input. | Low      |

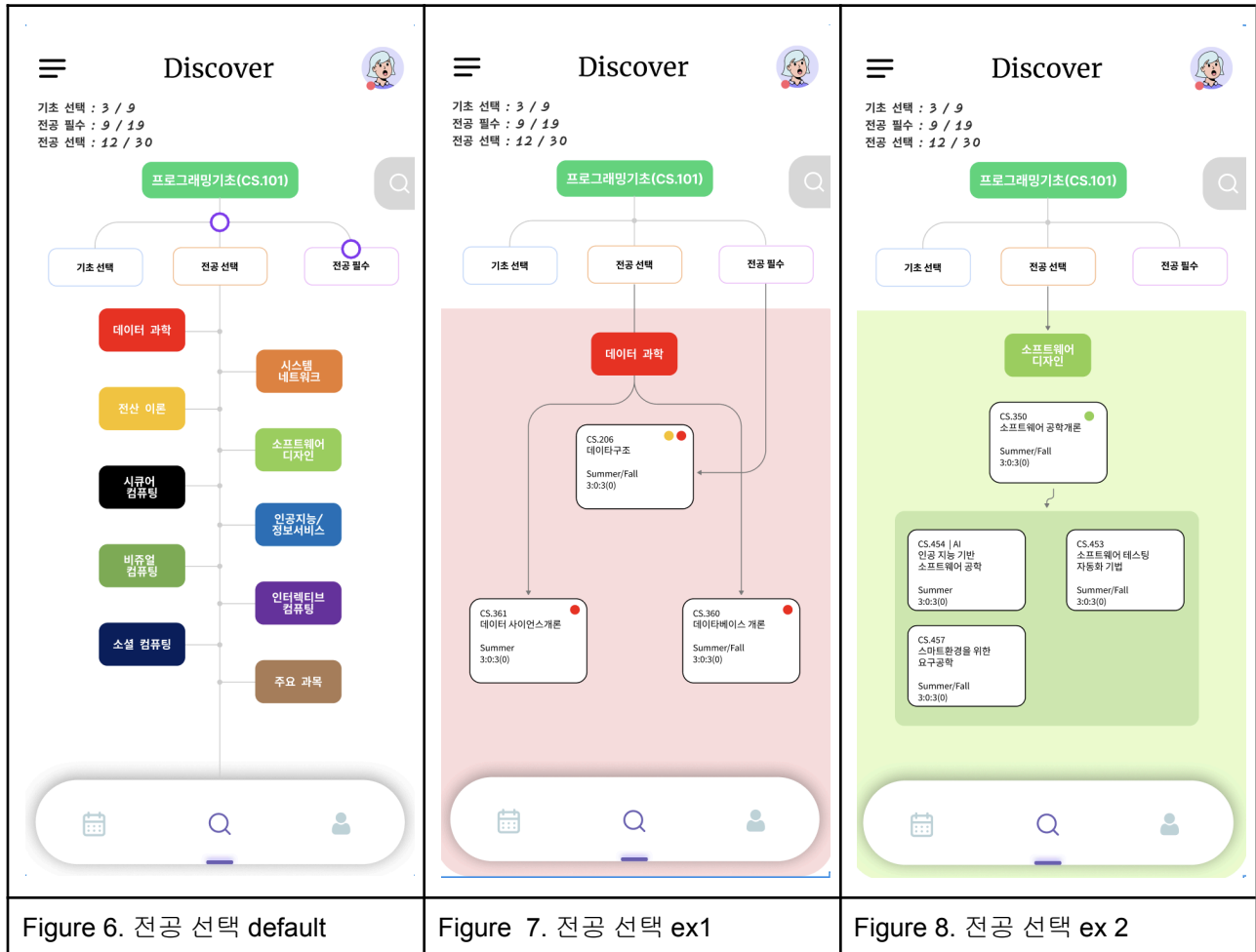
The use case diagram for the Calculator module is shown below.



## 4.2 Course Prerequisite Tree

### 4.2.1 Description and Priority





The main screen of the KAIST CS Roadmap Planner is the Course Prerequisite Tree module that visualizes the Computer Science curriculum at KAIST. The root node is Programming Basics (CS101), which branches into three primary categories: General Elective, Major Elective, and Major Required.

When a user clicks on any of these category nodes, the corresponding courses are expanded below in a hierarchical tree structure. Each course is displayed as a card. The card includes the following information: course code, course name, offered semester (Spring/Fall), and credit structure (Lecture:Lab:Credit). Additionally, each card contains a colored circular indicator at the top-right corner, representing the category or sector within the Computer Science curriculum that the course belongs to.

For example, when the General Elective node is selected, three courses are displayed below. Among them, Programming Lab is marked with a yellow circle, indicating that it belongs to the theoretical CS sector. When the Major Required node is selected, the tree structure also includes connections from Major Elective. This is because some required courses have Prerequisite courses that belong to the Major Elective category. Therefore, even when viewing Major Required courses, the tree maintains Prerequisite relationships across categories.

Furthermore, key courses within the curriculum are visually grouped with an additional brown boundary box to emphasize their importance. Courses that have already been marked as

Completed by the user are visually distinguished (e.g., grayed out), as shown in the System Programming course card in Figure 2. At the top-left corner of the screen, a mini status panel displays the number of Completed and required credits for each category (General Elective, Major Required, Major Elective), allowing users to quickly track their academic progress.

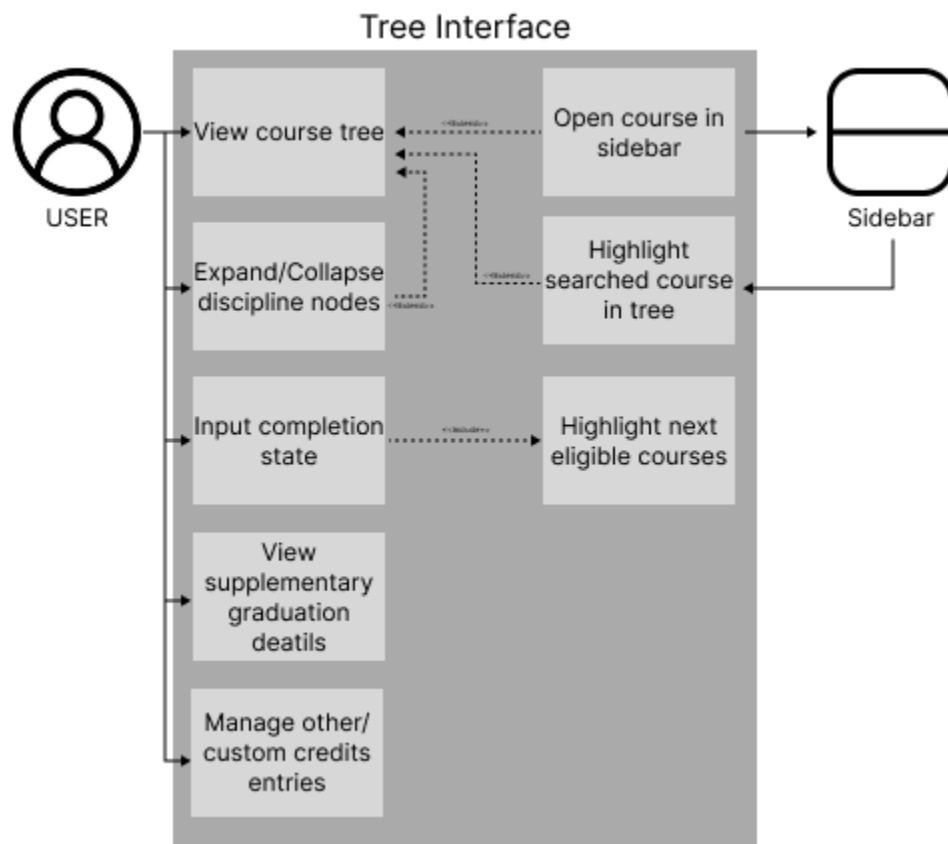
#### 4.2.2 Stimulus/Response Sequences

- User opens the app (Course Prerequisite Tree is the default/main view) → System renders the top-level tree with General Required and major category nodes visible
- User clicks a category node (e.g., Data Science, CS Theory) → System expands the subtree showing only courses within that discipline
- User clicks on an individual course node → System updates the real-time credit counter at the top and marks the course
- User searches for a course → System filters the Course Prerequisite Tree and shows the course's level and approximate position in the hierarchy

#### 4.2.3 Functional Requirements

| ID                  | Requirement Description                                                                                                                                                                                                                                       | Priority |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>REQ-TREE-001</b> | The system shall render the full course Prerequisite hierarchy as a collapsible tree. The initial view shall show only the top-level General Required courses and major category group nodes.                                                                 | High     |
| <b>REQ-TREE-002</b> | Each major discipline (e.g., AI, Systems-Networking, Theory, Software, etc.) shall be represented as a collapsible sub-tree node. Clicking a discipline node shall expand or collapse the courses within it.                                                  | High     |
| <b>REQ-TREE-003</b> | The tree shall visually differentiate course types: Major Required, Major Elective, Graduation Research, and Custom entries.                                                                                                                                  | High     |
| <b>REQ-TREE-004</b> | Clicking an individual course node shall open the course details in the Sidebar. Users shall be able to mark the course as Planned from the Sidebar, and the running credit counter in the Course Prerequisite Tree view shall update in real time.           | High     |
| <b>REQ-TREE-005</b> | The system shall provide a text search field (per OTL search UI conventions) that filters the Course Prerequisite Tree to matching courses and displays each result's approximate level (100/200/300/400-level) and its location in the hierarchy.            | High     |
| <b>REQ-TREE-006</b> | Completed courses shall be visually marked and the Course Prerequisite Tree shall highlight or prioritize courses the student is eligible to take next.                                                                                                       | High     |
| <b>REQ-TREE-007</b> | Graduation Research shall be embedded as a node within the Course Prerequisite Tree rather than treated as a separate module.                                                                                                                                 | Medium   |
| <b>REQ-TREE-008</b> | The Course Prerequisite Tree shall include a 기타 (Other) tab/section for non-standard credit recognition: internships, independent study (개별연구), substitute courses (대체과목), and other approved credit sources. These shall appear as Custom placeholder nodes. | Medium   |
| <b>REQ-TREE-009</b> | When switched to 'minor' mode, the Course Prerequisite Tree shall update to display the minor program's course hierarchy.                                                                                                                                     | Medium   |

The use case diagram for the Tree module is shown below.



## 4.3 Semester Roadmap

### 4.3.1 Description and Priority

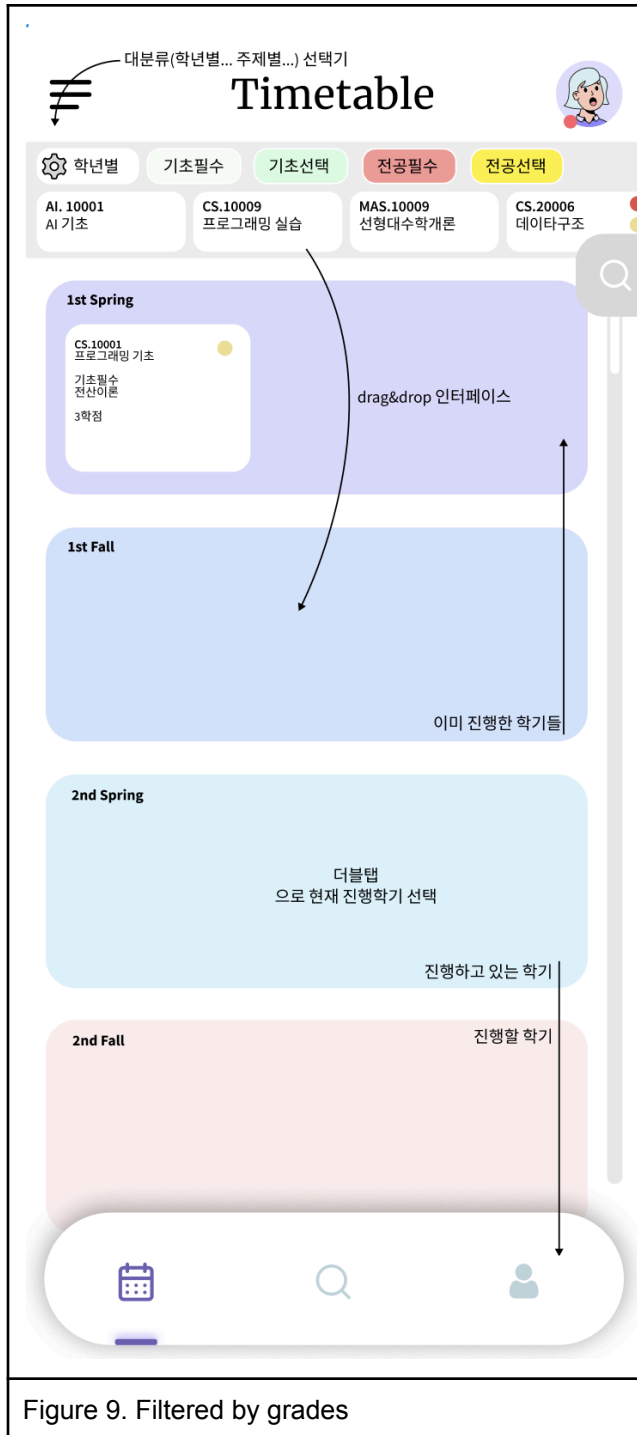
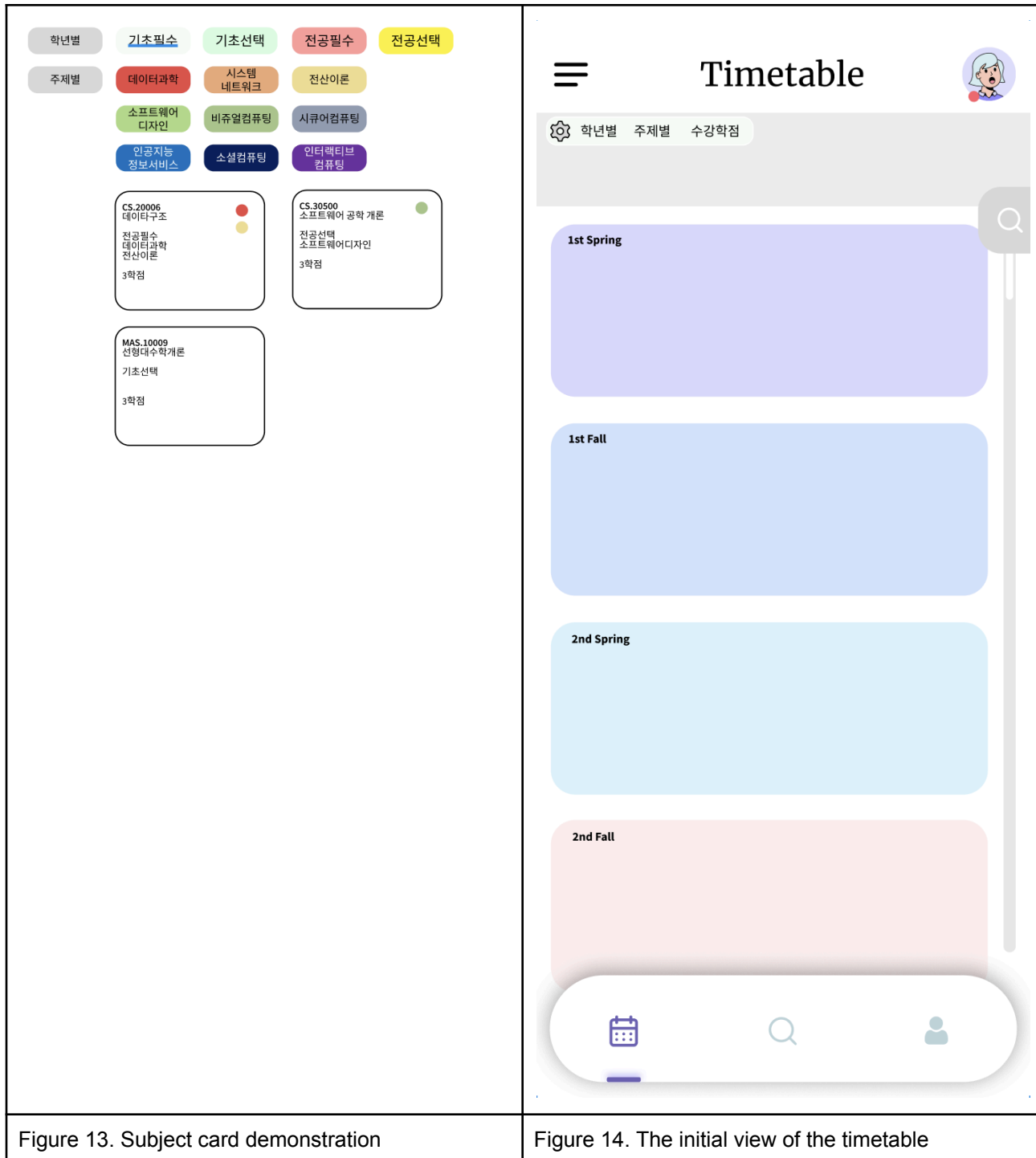




Figure 11. Required credits



Figure 12. Filter selector



NOTE: the arrows associated with letters shall NOT be on the running version of the app, it is a guidance for the explanations below.

The Semester Roadmap module provides a comprehensive multi-year view of the student's Planned schedule, utilizing subject cards that vary in detail (small vs. big) based on their display state. The visual system employs a dual-color coding scheme: a base color determined by grade



category (General Required, General Elective, Major Required, Major Elective) and sub-colors reflecting specific subtopics such as Data Science or Systems.

A sticky top UI displays subjects selected from the Sidebar or Course Prerequisite Tree, sorted numerically by their subject codes (excluding department prefixes). This interface includes a filter selector (cog icon) that allows users to toggle filters by grade, subtopic, or credits.

The layout consists of semester rows that support both individual horizontal scrolling and global vertical scrolling, with an option to add additional semesters at the bottom. Core functionalities include Drag&Drop for moving cards between the top UI and semester rows, and a Double Tap interaction to designate a "current semester," which triggers visual distinctions for past, current, and future academic periods.

#### 4.3.2 Stimulus/Response Sequences

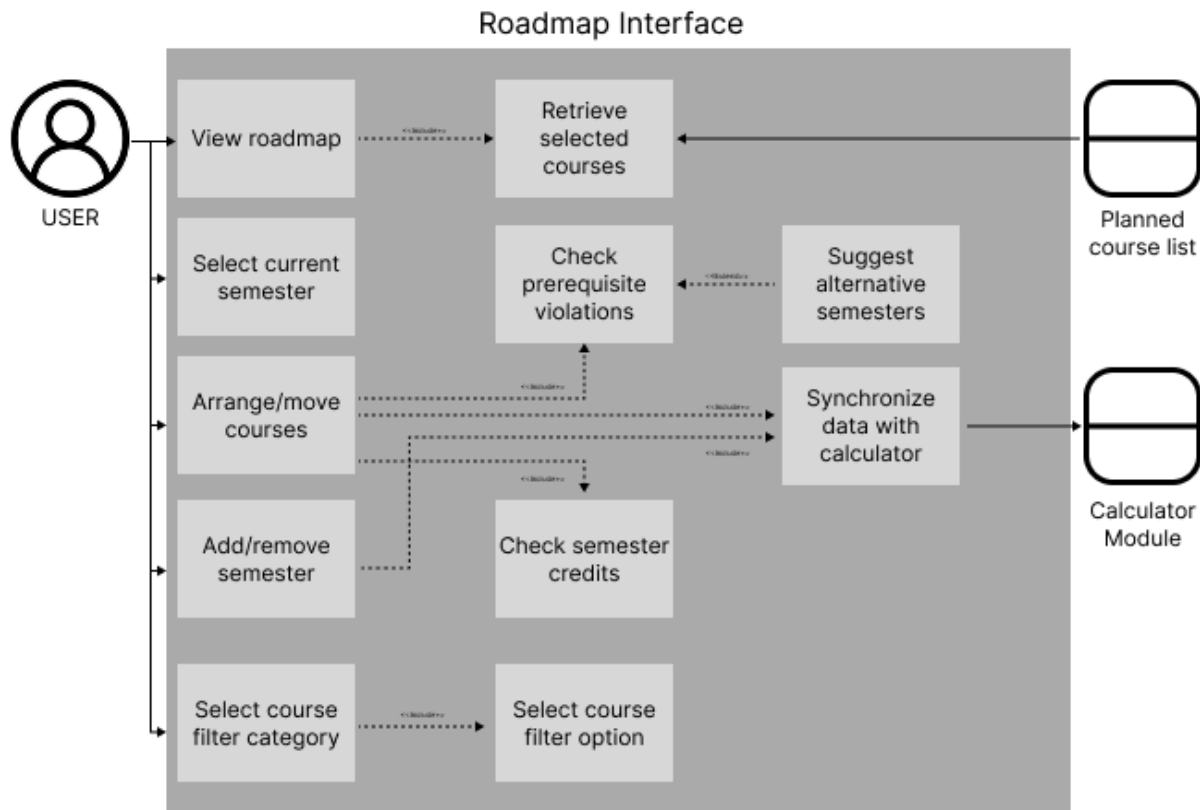
- User opens Semester Roadmap tab → System displays a grid of semester rows (e.g., 1-1 through 4-2), each containing course cards
- User drags a course card from one semester to another → System updates the course's semester assignment and recalculates all counters
- User clicks 'Add Semester' (+) button → System appends a new empty semester row to the grid
- User switches filter (annotated as cog icon on top of screen) → System filters the roadmap to show only courses relevant to the selected track
- User drags a course from the Course Prerequisite Tree into the Semester Roadmap → System places it in the target semester and marks it as Planned
- Double taps the semester row → System sets that row as 'current semester', also marking the previous semester as Completed

#### 4.3.3 Functional Requirements

| ID                 | Requirement Description                                                                                                                                                                                                                    | Priority |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>REQ-MAP-001</b> | KAIST CS Roadmap Planner shall display a scrollable semester grid where each row represents one academic semester (e.g., Spring 1, Fall 1, ... Spring 4, Fall 4).                                                                          | High     |
| <b>REQ-MAP-002</b> | Each semester row shall display the list of courses Planned for that semester as draggable cards.                                                                                                                                          | High     |
| <b>REQ-MAP-003</b> | The system shall support drag-and-drop reordering: users shall be able to drag course cards between semester rows. The top UI containing unplaced courses shall be fixed/sticky while the semester rows scroll vertically or horizontally. | High     |
| <b>REQ-MAP-004</b> | The system shall display a credit counter per semester row showing total credits for planned courses in that semester.                                                                                                                     | High     |
| <b>REQ-MAP-005</b> | Product shall support a double tap action on the screen, to implement the function 'selecting the current semester'.                                                                                                                       | High     |
| <b>REQ-MAP-006</b> | Courses shall be visually categorized using color coding or badges corresponding to their type: General Required, General Elective, Major Required, Major Elective, and Graduation Research.                                               | Medium   |

| ID                 | Requirement Description                                                                                                                                                                                      | Priority |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>REQ-MAP-007</b> | The system shall warn users (visual indicator) when a course is placed in a semester before its prerequisite is completed, based on the user's current roadmap state.                                        | Medium   |
| <b>REQ-MAP-008</b> | The system shall support an 'Add Semester' (+) button at the bottom of the pre-existing rows that appends a new empty semester row to accommodate students who take a leave of absence or extended programs. | Medium   |
| <b>REQ-MAP-009</b> | The roadmap shall support two grouping modes: grouping by academic year (year 1, year 2, etc.) and by subject area/topic.                                                                                    | Low      |
| <b>REQ-MAP-010</b> | Major Required courses that cannot be placed in the current semester due to schedule conflicts (as noted by the user) shall be highlighted and the system shall suggest alternate semesters.                 | Low      |
| <b>REQ-MAP-011</b> | The roadmap shall be synchronized with the Credit and GPA Calculator — credits for Planned courses in the roadmap shall be reflected in the calculator's projected totals.                                   | High     |
| <b>REQ-MAP-012</b> | The roadmap shall get data from the database to display the user's planned subjects.                                                                                                                         | High     |

The use case diagram for the Roadmap module is shown below.



## 4.4 Sidebar

### 4.4.1 Description and Priority



Figure 15. Default sidebar

Figure 16. Searched with categories

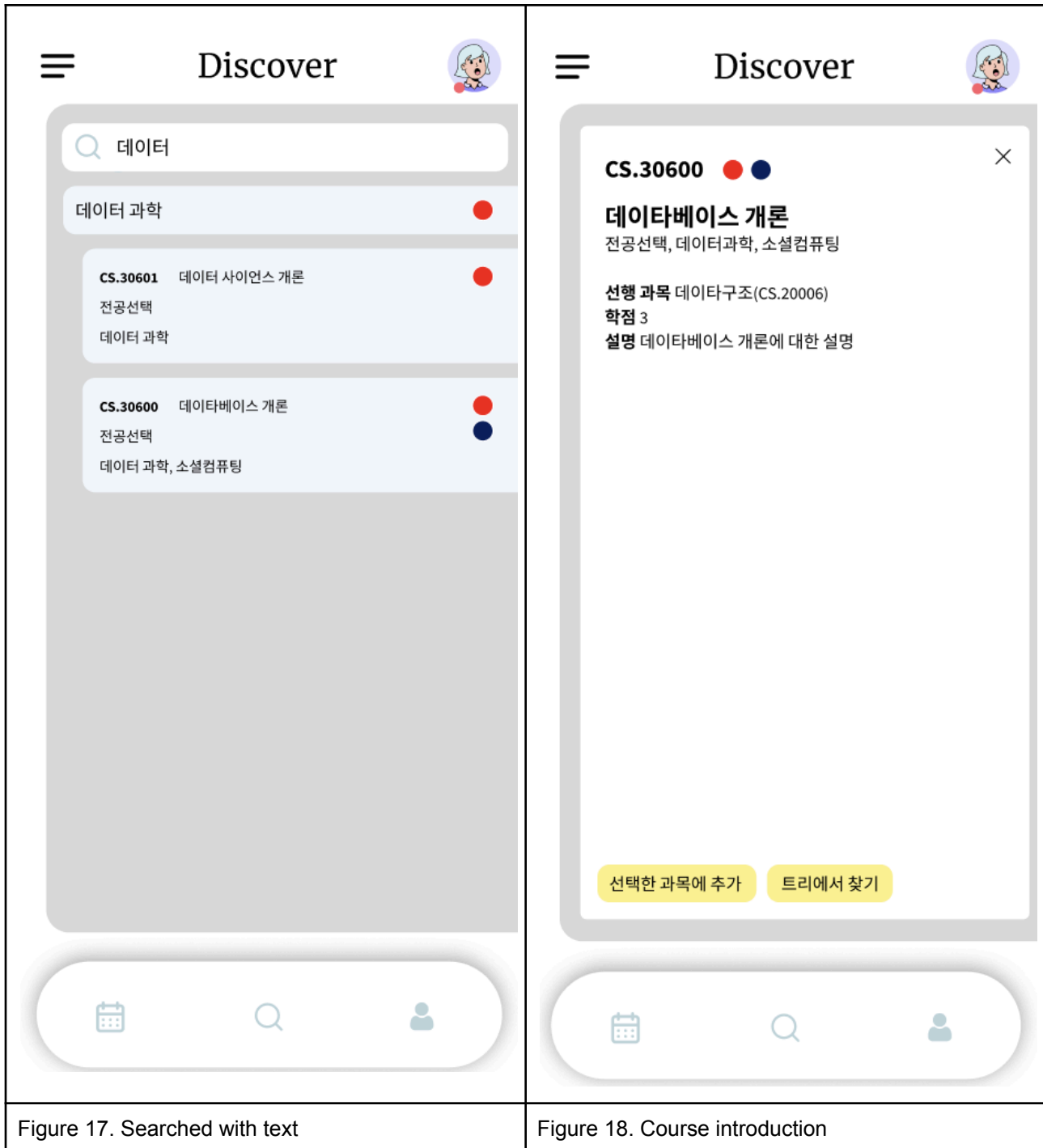
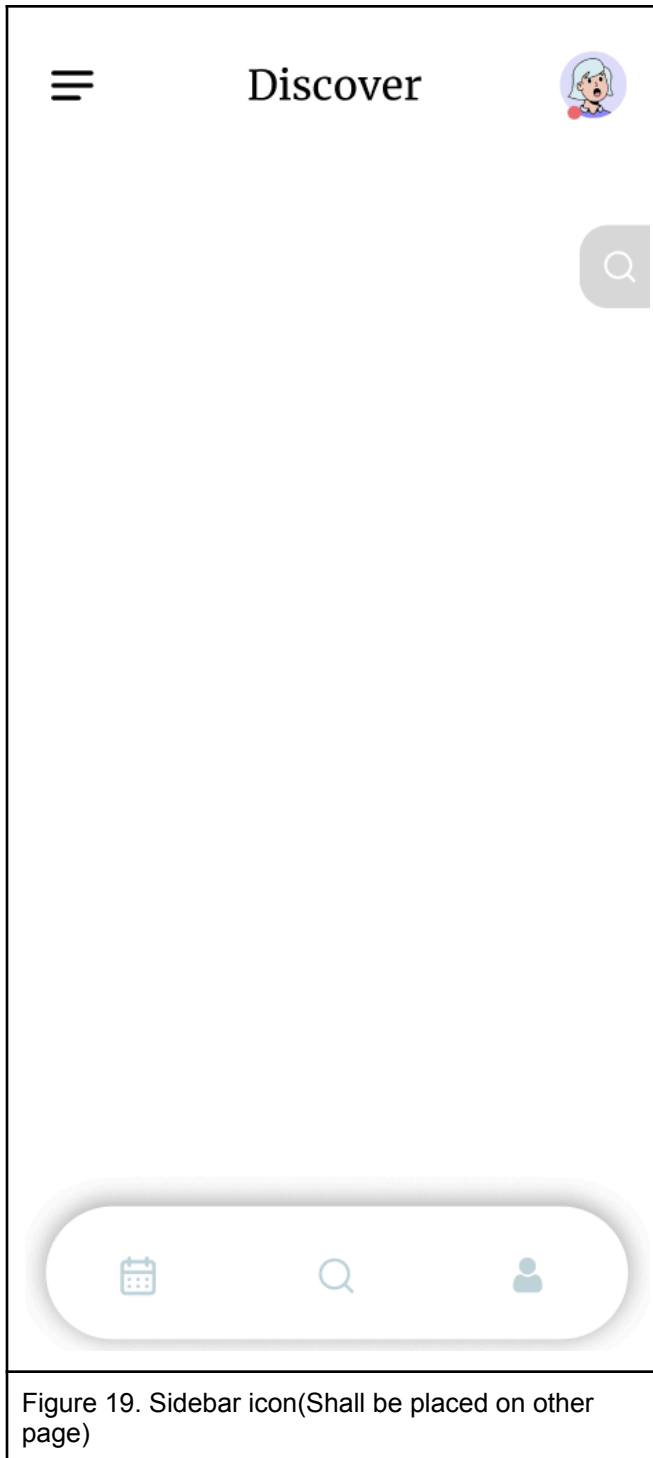


Figure 17. Searched with text

Figure 18. Course introduction



The sidebar is persistent across all tabs and supports open/close functionality. It enables subject searches via categories and text input. Upon selection, the sidebar provides subject-specific descriptions and actions. In the subject-specific page, the system shall show “Add to Planned Subjects” button and “Find in tree” button. Also, the cancel and go back function in the sidebar shall be offered.

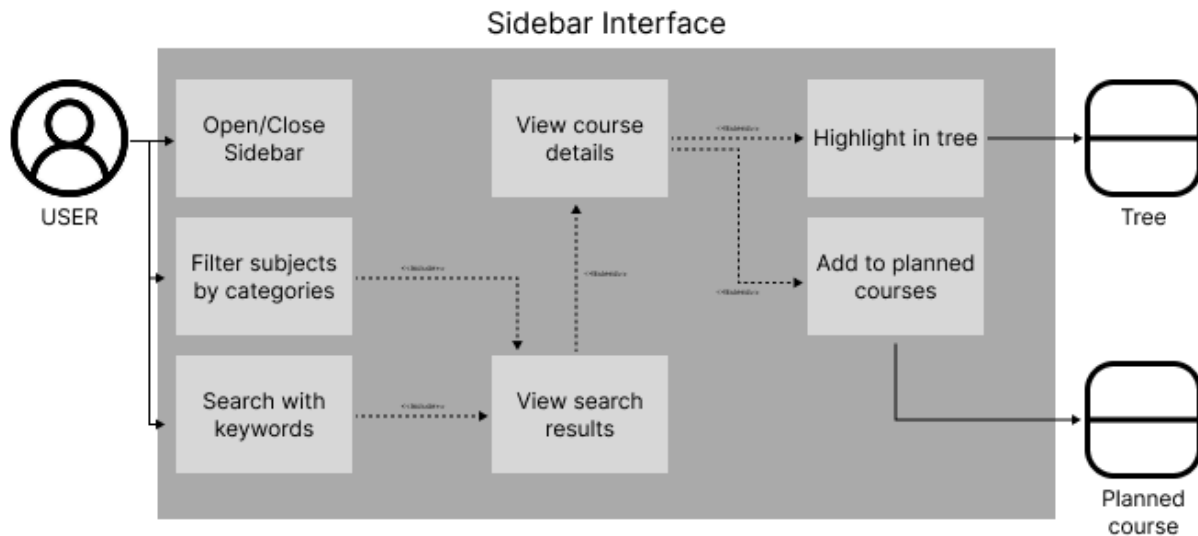
#### 4.4.2 Stimulus/Response Sequences

- User clicks the sidebar icon (located in the top-right corner of any tab) → System expands the sidebar and displays available subject categories
- User selects a category or enters text into the search bar → System filters and displays a list of matching subjects within the sidebar
- User clicks a specific subject → System displays the subject's detailed information and two buttons: 'Find in Tree' and 'Add to Planned Subjects'
- User clicks 'Find in Tree' → System navigates to and highlights the corresponding subject's location within the prerequisite tree structure
- User clicks 'Add to Planned Subjects' → System adds the chosen subject to the user's current selection list

#### 4.4.3 Functional Requirements

| ID                   | Requirement Description                                                                                           | Priority |
|----------------------|-------------------------------------------------------------------------------------------------------------------|----------|
| <b>REQ-SBA R-001</b> | Clicking the sidebar icon shall expand the sidebar to reveal the search interface and categories.                 | High     |
| <b>REQ-SBA R-002</b> | The system shall provide a function to close the expanded sidebar, returning it to the icon state.                | High     |
| <b>REQ-SBA R-003</b> | The system shall display a list of predefined categories to allow users to filter subjects by field.              | High     |
| <b>REQ-SBA R-004</b> | The system shall provide a search bar at the top of the sidebar for keyword-based text searches.                  | High     |
| <b>REQ-SBA R-005</b> | Upon selecting a subject from the results, the system shall display the subject's description and action buttons. | Medium   |
| <b>REQ-SBA R-006</b> | The 'Find in Tree' button shall navigate to and highlight the subject within the prerequisite tree.               | Medium   |
| <b>REQ-SBA R-007</b> | The 'Add to Planned Subjects' button shall include the chosen subject in the user's planning list.                | Medium   |

The use case diagram for the Sidebar module is shown below.



## 4.5 Settings

### 4.5.1 Description and Priority

The Settings module is responsible for managing the overall user configuration and academic profile within the KAIST CS Roadmap Planner. Users can personalize their interface through theme selections and define their academic status, such as Major, Double Major, or Minor. This data is synchronized with the Course Prerequisite Tree, Semester Roadmap, and Credit and GPA Calculator to accurately reflect and display requirements.

### 4.5.2 Stimulus/Response Sequences

- User navigates to the Settings page → System displays the current configuration options including theme, language, major/minor selection, and reset option
- User toggles the dark/light mode switch → System immediately applies the selected theme across the interface and persists the preference for future sessions
- User selects a language preference (Korean / English) → System immediately updates all interface text to the selected language and persists the preference across sessions
- User selects their Major program → System saves the selection and updates the Course Prerequisite Tree, Semester Roadmap, and Credit and GPA Calculator to reflect the corresponding requirements
- User optionally selects a Double Major or Minor program → System saves the selection and updates the relevant views to reflect the additional program requirements alongside the primary major
- User clicks the reset option → System displays a confirmation prompt warning that all course data will be cleared
- User confirms the reset action → System clears all user course data and restores the planner to its default state

### 4.5.3 Functional Requirements

| ID          | Requirement Description                                                                                                                                        | Priority |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| REQ-SET-001 | The system shall provide a sound toggle in the Settings page for enabling or disabling notification sound. The selected state shall persist across sessions.   | Medium   |
| REQ-SET-002 | The system shall provide a vibration toggle in the Settings page for enabling or disabling vibration alerts. The selected state shall persist across sessions. | Medium   |
| REQ-SET-003 | The system shall allow users to select their academic track, including major, minor, or double major, through the Settings page.                               | High     |
| REQ-SET-004 | The selected academic track shall affect the curriculum structure and related requirement views shown in the Tree, Roadmap, and Calculator modules.            | High     |
| REQ-SET-005 | The Settings page shall display the currently selected academic track in a clear and editable form, such as a dropdown selector.                               | Medium   |

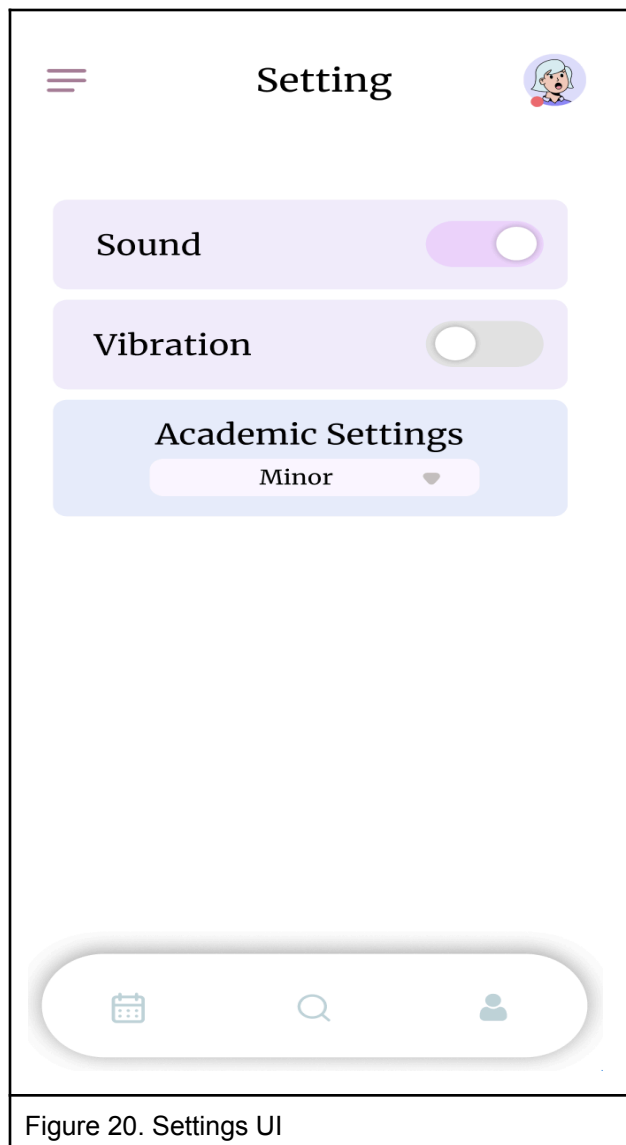


Figure 20. Settings UI



## 4.6 Authentication

### 4.6.1 Description and Priority

Handles authentication to secure and synchronize Completed and Planned records across devices. It protects personalized Course Prerequisite Tree and Credit and GPA Calculator data, supporting seamless access through KAIST SSO integration.

### 4.6.2 Stimulus/Response Sequences

- User opens the app without an active session → System displays the Login page with a 'Sign in with KAIST' button
- User clicks 'Sign in with KAIST' → System initiates the KAIST SSO redirect and displays a branded loading/splash screen during authentication processing
- User completes successful SSO authentication → System redirects the user to the Tree (main) view with their personalized data loaded
- User opens the app with an existing active session → System detects the valid session token and automatically redirects the user to the Tree (main) view without prompting for login
- SSO authentication fails → System displays a localized error message on the Login page along with a retry option
- User clicks the retry option → System re-initiates the KAIST SSO flow
- User selects the sign-out action → System invalidates the current session token and redirects the user to the Login page

### 4.6.3 Functional Requirements

| ID                  | Requirement Description                                                                                                                       | Priority |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>REQ-AUTH-001</b> | The system shall present a Login page with a 'Sign in with KAIST' button that initiates the KAIST SSO flow.                                   | High     |
| <b>REQ-AUTH-002</b> | The system shall display a branded loading/splash screen during SSO redirect and authentication processing.                                   | Medium   |
| <b>REQ-AUTH-003</b> | Upon successful authentication, the system shall redirect the user to the Tree (main) view.                                                   | High     |
| <b>REQ-AUTH-004</b> | The system shall maintain session state with an appropriate token lifetime; users shall not need to re-login within a single browser session. | High     |
| <b>REQ-AUTH-005</b> | The system shall handle SSO errors gracefully and display a localized error message with a retry option.                                      | Medium   |
| <b>REQ-AUTH-006</b> | A sign-out action shall invalidate the session and return the user to the Login page.                                                         | High     |

## 5. Other Nonfunctional Requirements

### 5.1 Performance Requirements

This application requires the following for performance. The Tree view initial render (with full course data loaded) shall complete within 2 seconds on a standard broadband connection. Drag-and-drop interactions shall have a response latency below 100ms to avoid perceived lag. GPA recalculation on course input shall be instantaneous ( $< 50\text{ms}$ , client-side computation). The application shall support at least 100 concurrent users without degradation, optimally 2 seconds in authentication and 1 second in data synchronization, in response time.

- Most planning interactions, including tree expansion, roadmap editing, and GPA updates, are assumed to be processed **client-side**.
- Server performance requirements primarily apply to **authentication** and **data synchronization**, rather than everyday planning interactions.

### 5.2 Safety Requirements

No safety-critical operations are involved — the system handles academic planning data only, with no financial transactions, medical decisions, or physical control implications. The following minimal safeguards apply:

- The system must prompt users for confirmation before any destructive action (e.g., resetting all course data via Settings).
- The system must not silently discard unsaved roadmap changes; it must either auto-save or warn the user before session expiry.
- Graduation requirement progress displayed in the Calculator must be clearly labeled as a planning aid and not an official KAIST academic record.

### 5.3 Security Requirements

For building credits and ensuring that user's personal data would be protected from risk. The application requires following;

- **Authentication:** Authentications must have one of two measures. It can be done by using KAIST SSO (SAML 2.0 or OpenID Connect) or if SSO is at no avail, authenticate via email, which only accepts the KAIST official email domains (namely, kaist.ac.kr). The application must never store raw passwords or credentials, by hashing it by salt & pepper or measures equivalent to it.
- **Data isolation:** Users should be only allowed to make a single account for one email. There should be no explicit nor implicit methods to view other user's data. No student may access or modify another student's planning data.
- **Session management:** All API endpoints require a valid authenticated session token. Sessions must expire after a configurable idle timeout (default: 1 hour).
- **Transport security:** All client-server communication must use HTTPS (TLS 1.2 or higher). Authentication tokens must be transmitted only via secure, HttpOnly cookies or Authorization headers — never in URL parameters.
- **CSRF protection:** All state-mutating endpoints (course updates, roadmap saves, settings changes) must implement CSRF protection.

- **Privacy:** Student planning data must not be shared with third parties, used for advertising, or included in any analytics beyond aggregate anonymized usage statistics (e.g., "most-viewed courses").

## 5.4 Software Quality Attributes

The application should be fairly easy to use including first-timers. This application's ultimate objective is to help user's flexible planning for future roadmaps. The quality attributes could be said as following;

- **Usability:** A first-time user must be able to complete the two core tasks, adding a course to the Roadmap and reading their current GPA, without consulting any external documentation. The onboarding walkthrough must cover these tasks.
- **Reliability:** The system must maintain 95% uptime for all server-dependent features (SSO, data sync) during KAIST academic semesters (March–June and September–December). Core planning features (tree browsing, GPA calculation) must remain functional offline using cached data.
- **Maintainability:** Course data and graduation requirement totals must be updatable via a remote configuration or backend update — no app store resubmission should be required for routine curriculum changes.
- **Portability:** The app must function on iOS 15+ and Android 9+ (SDK version 28 or higher). If built with a cross-platform framework (for example, React Native, Flutter), UI and behavior must be consistent across both platforms with no platform-specific feature gaps in core functionality.
- **Accessibility:** The app must support system-level accessibility features or should be changed via app settings: Dynamic Type / font scaling on iOS, TalkBack on Android, and sufficient color contrast (WCAG 2.1 AA) for all text and interactive elements. Dark/light mode must follow the system setting by default.
- **Scalability:** The app architecture must allow adding new course categories or modules without a full rewrite of existing components.

## 5.5 Business Rules

To ensure the user to use the app without serious research or user documentations, the app should have following policies;

- **Graduation requirement authority:** The credit totals and category definitions (기필, 기선, 전필, 전선, 졸업연구) displayed in the Calculator must mirror the official KAIST School of Computing undergraduate curriculum document for the student's entry year. The app is a planning aid, not an authoritative academic record.
- **GPA scale:** All GPA calculations must use the KAIST 4.3-point scale (A+: 4.3, A0: 4.0, B+: 3.3, B0: 3.0, C+: 2.3, C0: 2.0, D+: 1.3, D0: 1.0, F: 0.0). No conversion to a 4.0 scale is provided.

## 6. Other Requirements

The following requirements do not fit neatly into prior categories but are necessary for a complete specification:

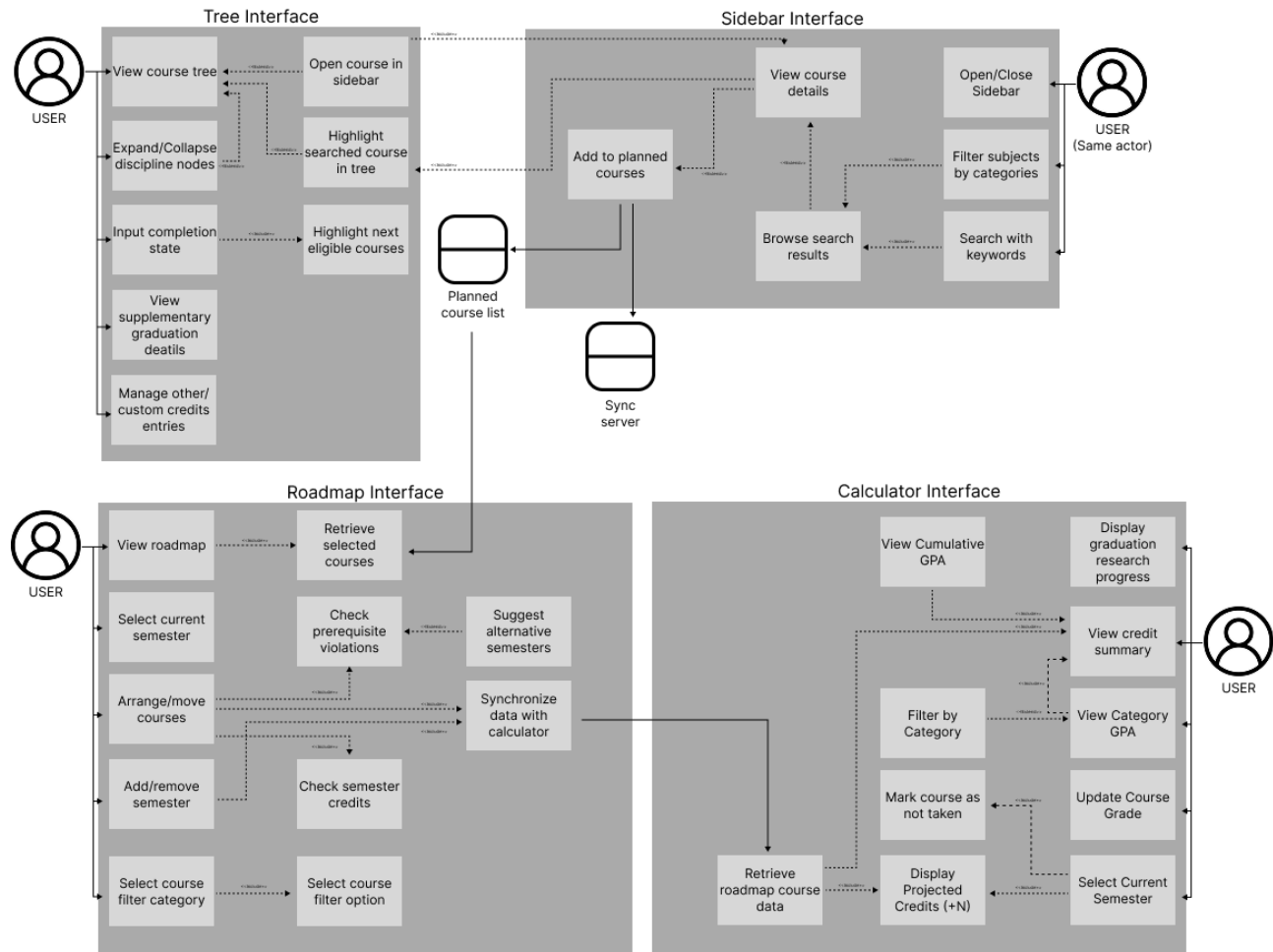
- **LOCALIZATION:** All primary UI labels and course category names (General Required, General Elective, Major Required, Major Elective, Graduation Research) shall be available in both Korean and English. The default language shall be Korean.
- **DATA PORTABILITY:** Users shall be able to export their Semester Roadmap and course selection as a JSON or CSV file.
- **ONBOARDING:** The system shall include a first-launch onboarding flow that explains the three main modules and guides users through entering their current completed courses.

## Appendix A: Glossary

| Term                               | Definition                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subtopic</b>                    | The subtopic of subjects, following the official KAIST CS Roadmap. (Includes 9 topics: Data Science, System-Network, Computational Theory, Software Design, Visual Computing, Secure Computing, AI-Information Service, Social Computing, Interactive Computing. For more detailed information, such as the course's topic, check the CS Roadmap on Reference page.) |
| <b>General Required</b> (기필, 기초필수) | General education required courses; compulsory for all CS students regardless of major track.                                                                                                                                                                                                                                                                        |
| <b>General Elective</b> (기선, 기초선택) | General education elective courses; chosen from a prescribed pool of general subjects.                                                                                                                                                                                                                                                                               |
| <b>Major Required</b> (전필, 전공필수)   | Major required courses; compulsory courses specific to the CS major.                                                                                                                                                                                                                                                                                                 |
| <b>Major Elective</b> (전선, 전공선택)   | Major elective courses; student-chosen courses within the CS major.                                                                                                                                                                                                                                                                                                  |
| <b>Graduation Research</b> (졸업연구)  | A capstone research requirement, typically worth 3 credits, required for graduation.                                                                                                                                                                                                                                                                                 |
| <b>OTL (Online Timetable)</b>      | KAIST official online course timetable system at otl.kaist.ac.kr.                                                                                                                                                                                                                                                                                                    |
| <b>SSO (Single Sign-On)</b>        | KAIST central authentication system used for all university services.                                                                                                                                                                                                                                                                                                |
| 인정학점                               | Approved/recognized credits for non-standard academic activities.                                                                                                                                                                                                                                                                                                    |
| 개별연구                               | Independent research credits, arranged directly with a faculty member.                                                                                                                                                                                                                                                                                               |

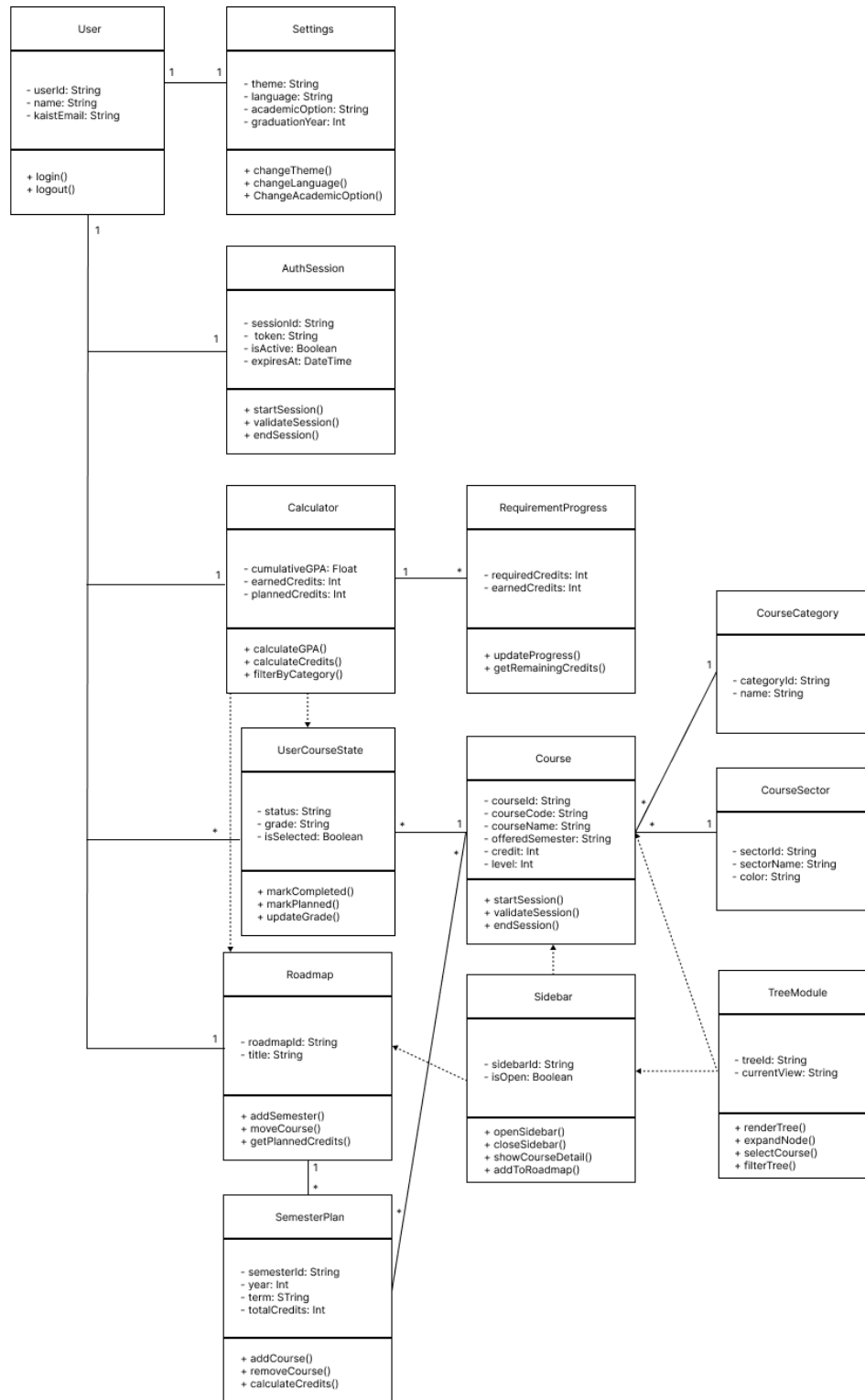
## Appendix B: Analysis Models

### B.1 Full Use Case Diagram



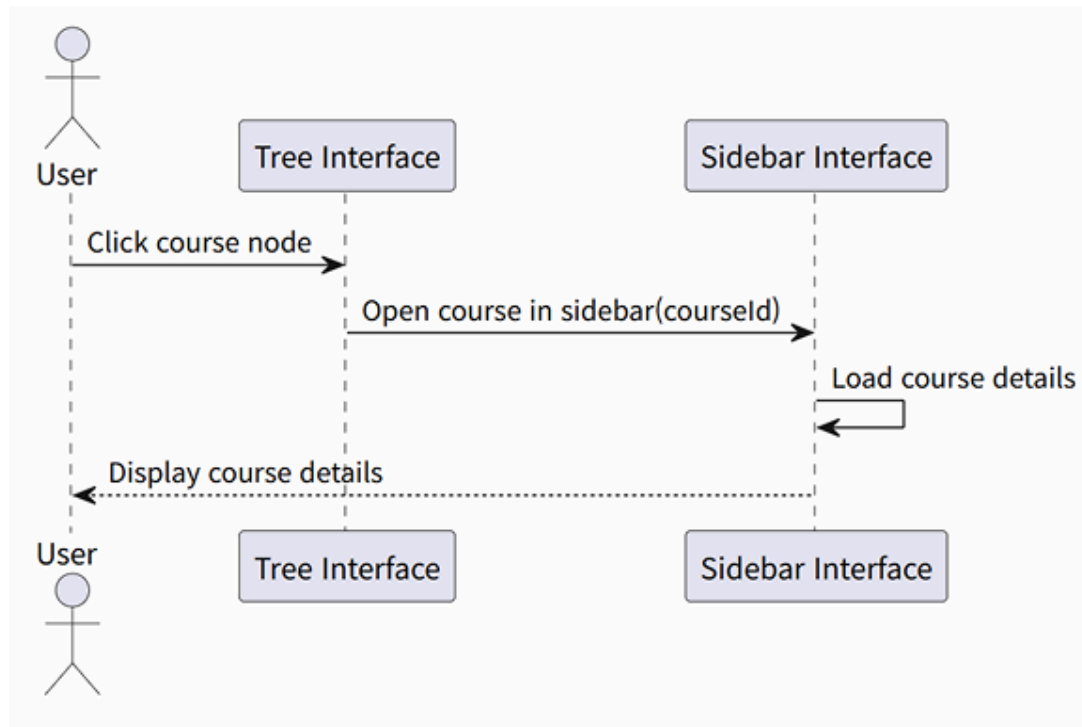
NOTE: Every USER actor in this use case diagram is the same actor.

## B.2 Class Diagram

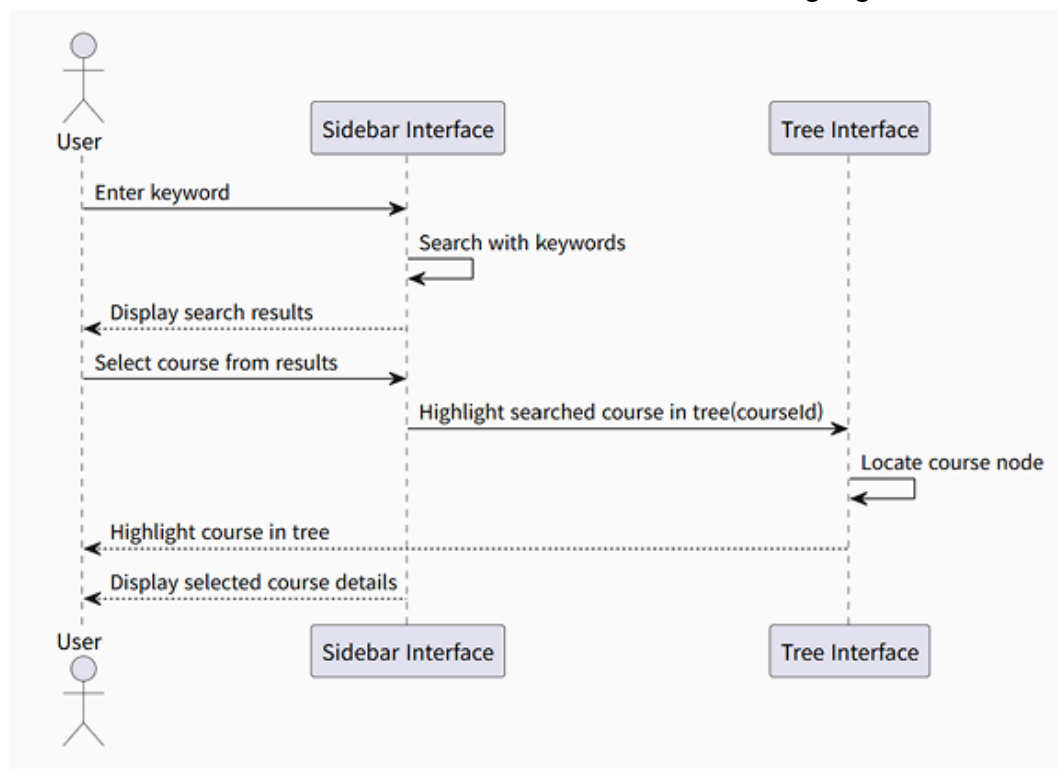


### B.3 Sequence Diagram

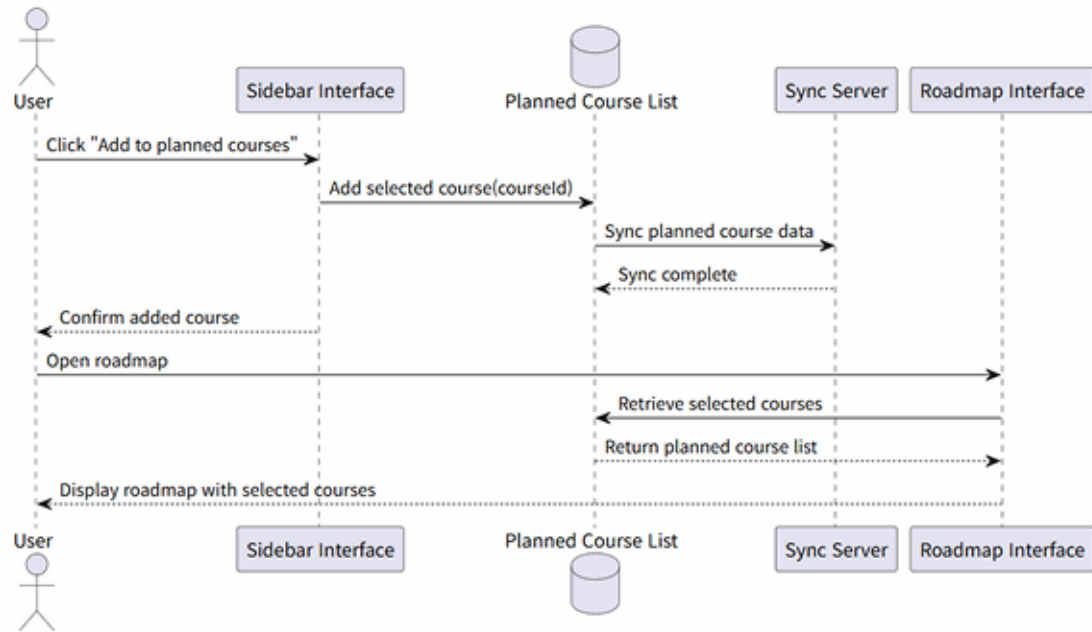
#### B.3.1 Select course from Tree → Sidebar shows detailed information



#### B.3.2 Search from Sidebar → Show results → Highlight Tree node

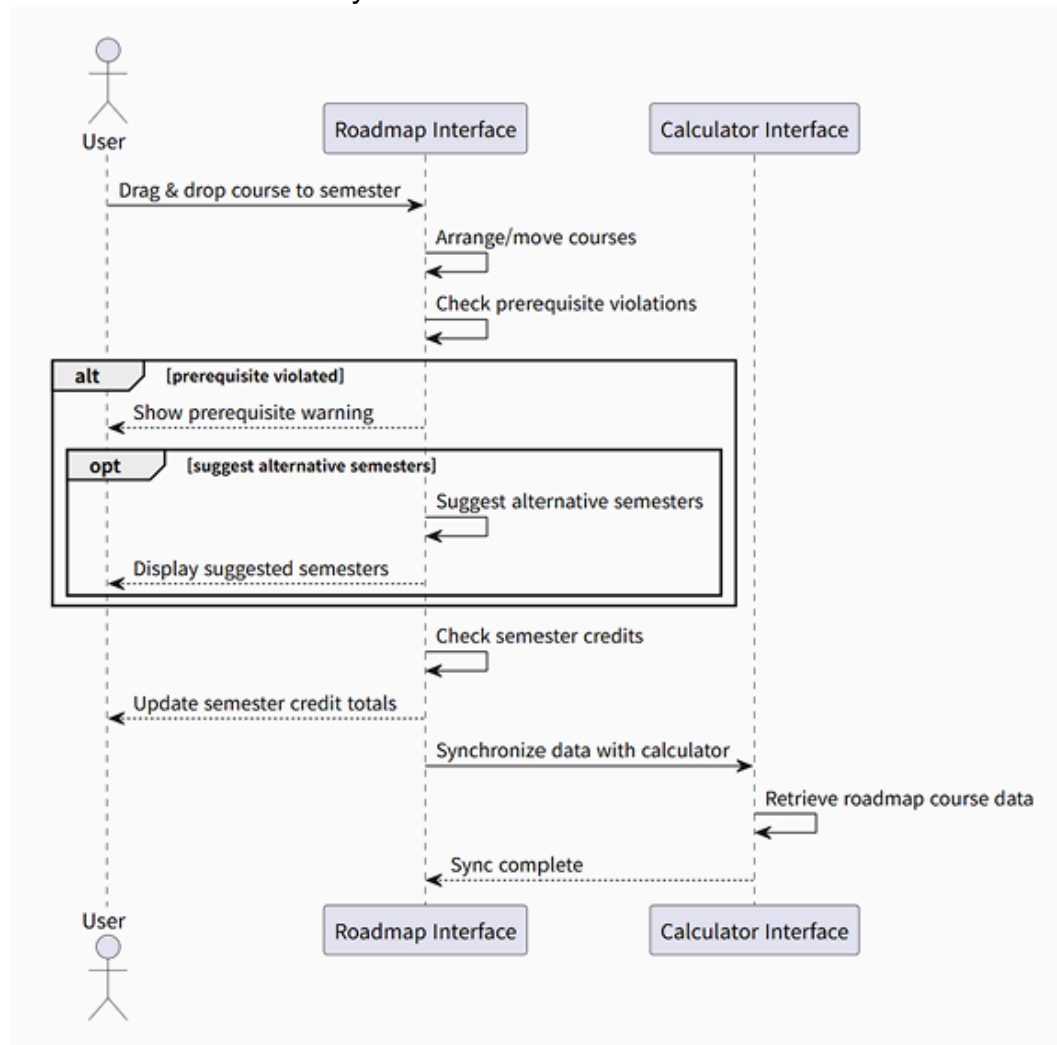


#### B.3.3 From Sidebar add to planned courses → Apply to Planned Course List → Retrieve at Roadmap

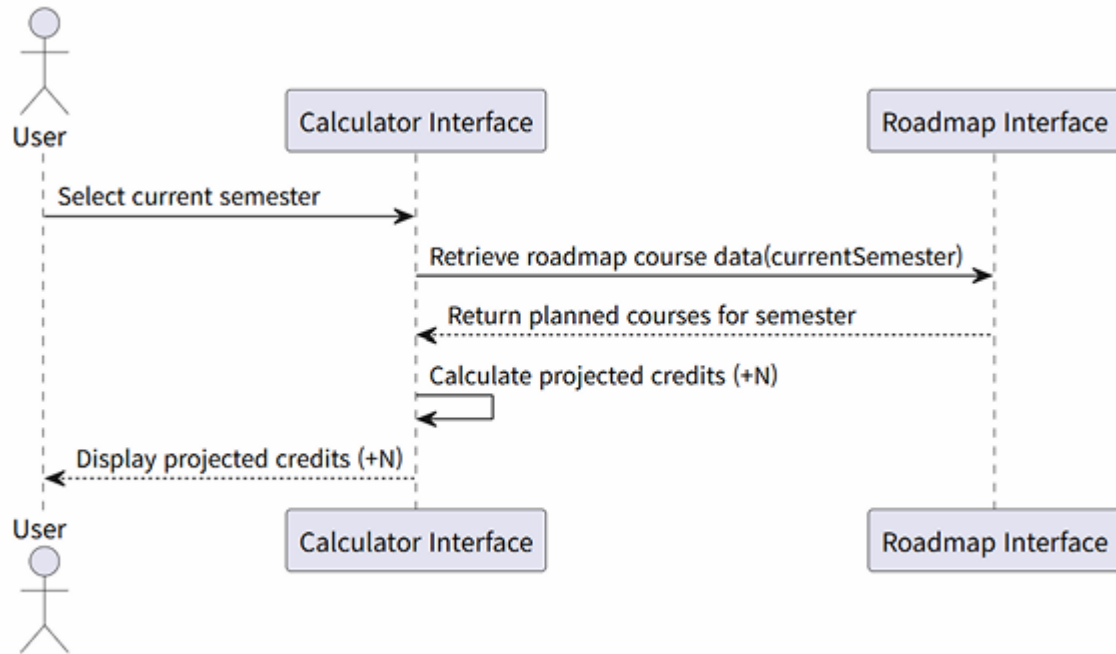




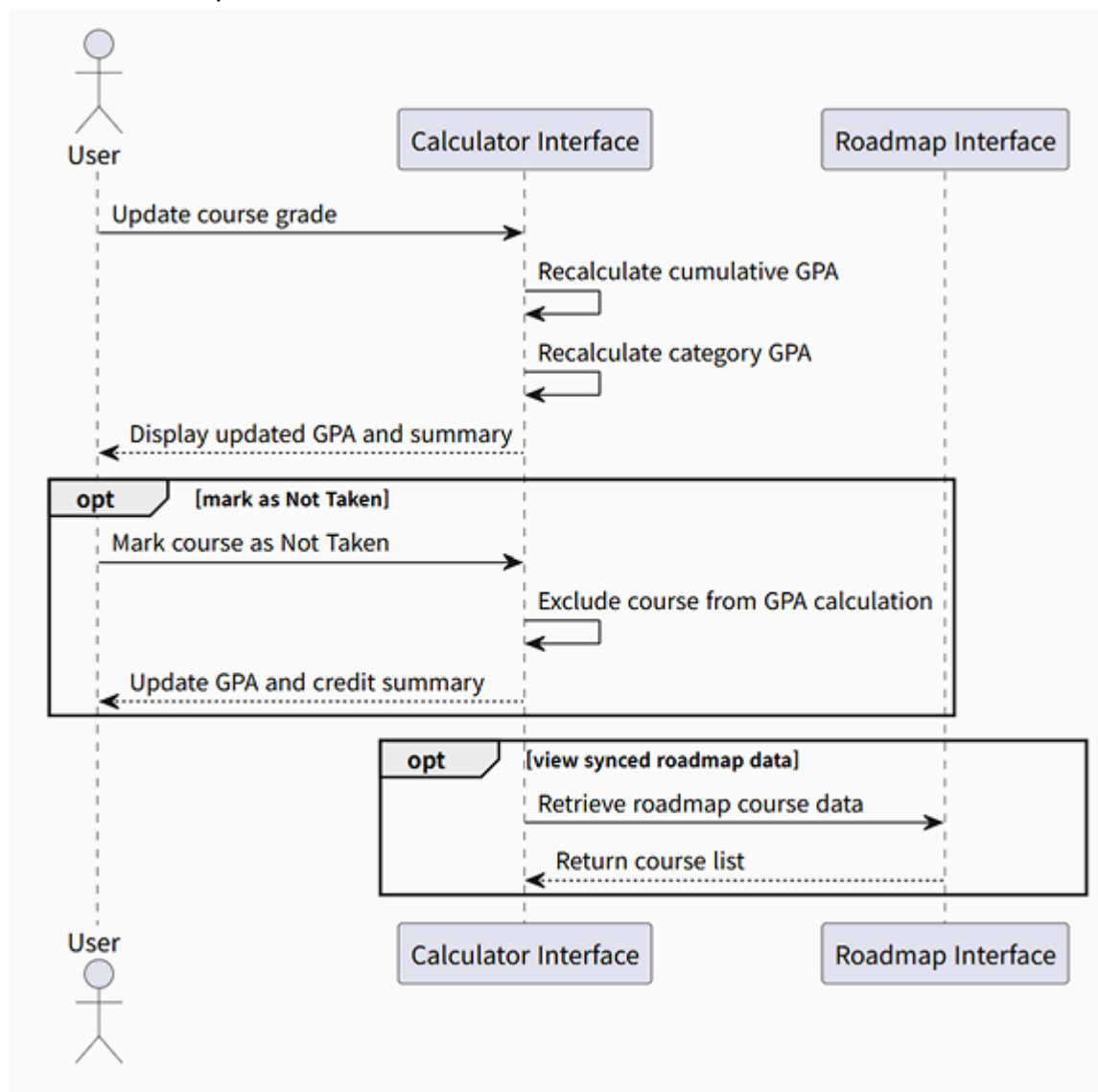
B.3.4 Move course card at Roadmap → Check prerequisite violations → GPA based on semester → Sync Calculator



## B.3.5 Select semester from Calculator → Apply Roadmap data → Show Projected Credits



## B.3.6 Input GPA in Calculator / Execute Not Taken / refresh GPA



## Appendix C: To Be Determined List

| ID      | Requirement Description                                                                                              | Priority |
|---------|----------------------------------------------------------------------------------------------------------------------|----------|
| TBD-001 | KAIST SSO integration method (SAML 2.0 vs. OpenID Connect) — pending confirmation from KAIST IT.                     | High     |
| TBD-002 | Official API or data feed availability from KAIST for course catalog and prerequisite data.                          | High     |
| TBD-003 | Exact graduation requirement credit counts per category for each entry-year cohort (2020–2024 curricula may differ). | High     |
| TBD-004 | Drag-and-drop library selection (e.g., dnd-kit, react-beautiful-dnd) — to be decided in technical design phase.      | Medium   |
| TBD-005 | Data persistence strategy: browser-only (IndexedDB) vs. cloud backend — v1.0 scope decision pending.                 | High     |
| TBD-006 | Recommended course list algorithm details — prerequisite-only or additional heuristics.                              | Low      |